

- RAG Learning

2.2 Formulae, Identities, Proof & Functions

AQA GCSE Mathematics 8300 — Algebra — spec refs A5–A7 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.2.1 Using Standard Formulae (A5)

A **formula** is a rule that links two or more variables. You need to be able to use formulae given in words or in symbols, including ones from science and other subjects. Some formulae must be learned (for example area and volume formulae); others, such as the equations of motion, are given in the question.

Subject of a formula: the single variable on its own on one side of the equals sign. In $A = lw$, A is the subject.

Substitute: replace each letter by its value, then evaluate using the correct order of operations.

Formulae for motion (given in the exam when needed), where u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement:

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as$$

Worked Example: Use $s = ut + \frac{1}{2}at^2$ to find s when $u = 3$, $t = 4$ and $a = 2.5$.

$$ut = 3 \times 4 = 12$$

$$\frac{1}{2}at^2 = \frac{1}{2} \times 2.5 \times 16 = 20 \text{ (square } t \text{ first)}$$

$$s = 12 + 20 = 32 \text{ m}$$

Worked Example: A formula in words: "To convert Celsius to Fahrenheit, multiply by 9, divide by 5, then add 32." Write this as a formula and convert 25°C .

$$F = \frac{9C}{5} + 32$$

$$F = \frac{9 \times 25}{5} + 32 = 45 + 32 = 77^\circ\text{F}$$

Worked Example: Use $v^2 = u^2 + 2as$ to find v when $u = 3$, $a = 4$ and $s = 2$ ($v > 0$).

$$v^2 = 9 + 2 \times 4 \times 2 = 9 + 16 = 25$$

$$v = \sqrt{25} = 5$$

Exam Tip: Write the formula, then write it again with the numbers substituted, before using a calculator. If you make a slip you can still earn the method mark.

2.2.2 Rearranging Formulae to Change the Subject (A5)

Rearranging a formula works exactly like solving an equation: do the **same inverse operation to both sides**. Undo the operations in the *reverse* order to the order in which they were applied to the new subject.

Operation on the subject	Undo it by...
$+ k$	subtracting k
$\times k$	dividing by k
$\div k$ (fraction)	multiplying by k

squaring	square-rooting (give $\pm$ unless told the value is positive)
square-rooting	squaring
cubing	cube-rooting

Worked Example: Make t the subject of $v = u + at$.

Subtract u : $v - u = at$

Divide by a : $t = \frac{v - u}{a}$

Worked Example: Make x the subject of $y = \frac{3x - 4}{5} + 2$.

Subtract 2: $y - 2 = \frac{3x - 4}{5}$

Multiply by 5: $5y - 10 = 3x - 4$

Add 4: $5y - 6 = 3x$

Divide by 3: $x = \frac{5y - 6}{3}$

Formulae with powers and roots

Worked Example: Make r the subject of $V = \frac{4}{3}\pi r^3$.

Multiply by 3: $3V = 4\pi r^3$

Divide by 4π : $r^3 = \frac{3V}{4\pi}$

Cube root: $r = \sqrt[3]{\frac{3V}{4\pi}}$

Worked Example: Make u the subject of $v^2 = u^2 + 2as$ (with $u > 0$).

Subtract $2as$: $u^2 = v^2 - 2as$

Square root: $u = \sqrt{v^2 - 2as}$

Exam Tip: $\sqrt{v^2 - 2as}$ is **not** the same as $v - \sqrt{2as}$. You cannot square-root a sum term by term. Keep the whole expression under one root sign.

2.2.3 When the Subject Appears Twice Higher

If the new subject appears in more than one term, you need four steps:

1. Clear any fractions (multiply both sides by the denominator) and expand brackets.
2. Collect all terms containing the subject on one side, everything else on the other.
3. **Factorise** the subject out as a common factor.
4. Divide by the bracket.

Worked Example: Make x the subject of $ax - 3 = bx + 7$.

Collect x terms: $ax - bx = 7 + 3$

Factorise: $x(a - b) = 10$

Divide: $x = \frac{10}{a - b}$

Worked Example: Make x the subject of $y = \frac{3x + 2}{x - 4}$.

Multiply by $(x - 4)$: $y(x - 4) = 3x + 2$

Expand: $xy - 4y = 3x + 2$

Collect x terms: $xy - 3x = 4y + 2$

Factorise: $x(y - 3) = 4y + 2$

Divide: $x = \frac{4y + 2}{y - 3}$

Exam Tip: Check by substituting a number. With $x = 5$: $y = \frac{17}{1} = 17$, and $\frac{4(17) + 2}{17 - 3} = \frac{70}{14} = 5 \checkmark$.

2.2.4 Equations and Identities (A6)

Equation: true only for particular value(s) of the letter. $2x + 1 = 9$ is only true when $x = 4$.

Identity: true for every value of the letter. Written with the symbol $\equiv$, read "is identically equal to". Example: $3(x + 2) \equiv 3x + 6$.

Showing two expressions are equivalent

To **show** an identity, start with one side (usually the more complicated one) and manipulate it until it becomes the other side. Show every step.

Worked Example: Show that $(x + 4)^2 - (x - 4)^2 \equiv 16x$. Hence work out $104^2 - 96^2$.

$$(x + 4)^2 = x^2 + 8x + 16 \text{ and } (x - 4)^2 = x^2 - 8x + 16$$

$$x^2 + 8x + 16 - x^2 + 8x - 16 = 16x \checkmark$$

$$\text{Hence with } x = 100: 104^2 - 96^2 = 16 \times 100 = 1600$$

Finding unknown constants in an identity

Because an identity is true for all x , the coefficients of each power of x must match on both sides ("compare coefficients").

Worked Example: $(x + a)(x + 3) \equiv x^2 + bx + 12$. Find a and b .

Expand: $x^2 + 3x + ax + 3a = x^2 + (3 + a)x + 3a$

Constants: $3a = 12 \Rightarrow a = 4$

x terms: $b = 3 + a = 7$

Exam Tip: To show a statement is *not* an identity, one **counter-example** is enough. For $(x + 2)^2 = x^2 + 4$, try $x = 1$: $9 \neq 5$.

2.2.5 Algebraic Argument and Proof (A6)

All students should be able to use algebra to support an argument. At Higher tier this extends to full **algebraic proof**, which must work for *every* case — testing examples is never a proof.

Useful general forms (n and m are integers):

Even number: $2n$ Odd number: $2n + 1$ Consecutive integers: $n, n + 1, n + 2$

Consecutive even numbers: $2n, 2n + 2$ Consecutive odd numbers: $2n + 1, 2n + 3$

Multiple of k : $k \times (\text{integer})$ Two *different* numbers: use different letters, e.g. $2m + 1$ and $2n + 1$

Worked Example: Show that the sum of any three consecutive integers is a multiple of 3.

Let the integers be $n, n + 1, n + 2$

Sum = $3n + 3 = 3(n + 1)$

$3(n + 1)$ is 3 times an integer, so it is always a multiple of 3.

Algebraic proof

Higher

Worked Example: Prove that the product of any two odd numbers is odd.

Let the odd numbers be $2m + 1$ and $2n + 1$

$$(2m + 1)(2n + 1) = 4mn + 2m + 2n + 1$$

$$= 2(2mn + m + n) + 1$$

$2(2mn + m + n)$ is even, so the product is (even) $+1$, which is odd.

Worked Example: Prove that $(2n + 3)^2 - (2n - 3)^2$ is a multiple of 12 for all positive integers n .

$$(4n^2 + 12n + 9) - (4n^2 - 12n + 9) = 24n$$

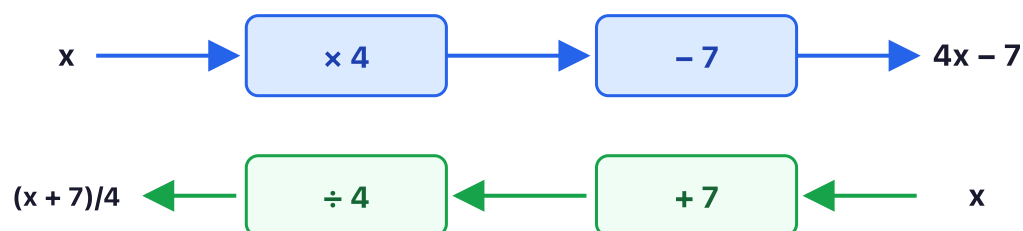
$24n = 12 \times 2n$, and $2n$ is an integer, so the expression is a multiple of 12.

Exam Tip: A proof needs a **concluding sentence** that links the algebra back to the claim (e.g. "...which is $3 \times$ an integer, so it is a multiple of 3"). Without it you usually lose the final mark.

2.2.6 Functions as Inputs and Outputs (A7)

A **function** is a rule that takes an input and gives exactly one output. Simple expressions can be thought of as functions, and function machines show the steps.

Function machine and its inverse



Worked Example: A function machine multiplies by 4 then subtracts 7. The output is 25. Find the input.

Work backwards, doing the inverse operations in reverse order

$$25 + 7 = 32, \text{ then } 32 \div 4 = 8$$

$$\text{Input} = 8 \text{ (check: } 8 \times 4 - 7 = 25 \checkmark)$$

2.2.7 Function Notation, Composite and Inverse Functions

Higher

$f(x)$ notation: $f(x) = 3x - 2$ means "the function f multiplies by 3 then subtracts 2".
 $f(4)$ means the output when the input is 4: $f(4) = 10$.

Composite function $fg(x)$: apply g first, then apply f to the result: $fg(x) = f(g(x))$.

Inverse function $f^{-1}(x)$: the function that reverses f , so $f^{-1}(f(x)) = x$.

Worked Example: $f(x) = 7 - 2x$. Find $f(-3)$ and solve $f(x) = 1$.

$$f(-3) = 7 - 2(-3) = 7 + 6 = 13$$

$$7 - 2x = 1 \Rightarrow 2x = 6 \Rightarrow x = 3$$

Composite functions

Worked Example: $f(x) = 2x - 5$ and $g(x) = x^2 + 1$. Find $fg(3)$ and $gf(x)$.

$$g(3) = 10, \text{ so } fg(3) = f(10) = 15$$

$$gf(x) = g(2x - 5) = (2x - 5)^2 + 1$$

$$= 4x^2 - 20x + 25 + 1 = 4x^2 - 20x + 26$$

Exam Tip: Order matters: $fg(x)$ and $gf(x)$ are usually different. Read $fg(x)$ from right to left — the function nearest the x acts first.

Inverse functions

Method: write $y = f(x)$, rearrange to make x the subject, then swap y for x .

Worked Example: Find $f^{-1}(x)$ when $f(x) = 3x + 2$.

$$y = 3x + 2$$

$$y - 2 = 3x \Rightarrow x = \frac{y - 2}{3}$$

$$f^{-1}(x) = \frac{x - 2}{3}$$

Worked Example: $f(x) = 3x - 2$ and $g(x) = x + 4$. Solve $f^{-1}(x) = g(x)$.

$$f^{-1}(x) = \frac{x + 2}{3}$$

$$\frac{x + 2}{3} = x + 4 \Rightarrow x + 2 = 3x + 12$$

$$-2x = 10 \Rightarrow x = -5$$

Exam Tip: $f^{-1}(x)$ is **not** $\frac{1}{f(x)}$. The $^{-1}$ here means “inverse function”, not “reciprocal”.

Quick Reference Summary

Skill	Key idea
Using formulae	Write formula, substitute, then evaluate in the correct order

Rearranging	Inverse operations to both sides, in reverse order
Powers and roots	Isolate the power first, then root ($\pm$ unless told positive)
Subject twice Higher	Collect, factorise, divide
Equation vs identity	Equation: some values; identity ($\equiv$): all values
Showing / proving	Use $2n, 2n + 1, n, n + 1...$; finish with a conclusion
Function machines	Reverse the operations to find an input
$fg(x)$ Higher	$f(g(x))$: g first, then f
$f^{-1}(x)$ Higher	$y = f(x)$, make x the subject, rename